FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2015**

HALF COURSE TEST –VI

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 09)** contains 09 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (10 – 13) contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (14 – 19) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (1 – 03)** contains 3 Matrix Match Type (4 × 4 Matrix) questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+6 marks** for all correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

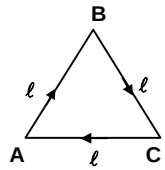
PART – I

SECTION – A

Single Correct Choice Type

*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

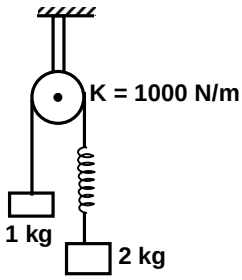
1. A particle moves over the sides of an equilateral triangle of side ℓ with constant speed v as shown in figure. The magnitude of average acceleration as it moves from A to C is



(A) $\frac{v^2}{\ell}$ (B) $\frac{\sqrt{3}}{2}\frac{v^2}{\ell}$

(C) $\frac{\sqrt{3}v^2}{\ell}$ (D) $\frac{v^2}{2\ell}$

2. In the arrangement shown in the figure pulley is light and smooth. The extension in the spring is ($g = 10\ m/s^2$)

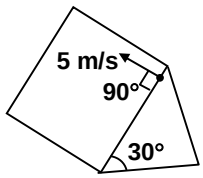


(A) 1.33 cm (B) 1 cm
(C) 1.67cm (D) 2 cm

3. A block of mass m is gently placed over a massive plank moving horizontally over a smooth surface with velocity 6 m/s. The coefficient of friction between the block and plank is 0.2. The distance travelled by the block till it stop with respect to the plank is ($g = 10\ m/s^2$)

(A) 4 m (B) 6 m
(C) 9 m (D) 12 m

4. A particle is given velocity 5 m/s on a fixed large inclined surface as shown in the figure. The radius of curvature of the path of the particle after 1 sec from the start is about($g = 10\ m/s^2$)



(A) 7 m (B) 14 m
(C) 21 m (D) 28 m

Space for rough work

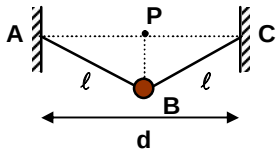
5. A liquid drop at temperature t, isolated from its surrounding breaks into a number of droplets the temperature of the droplets will be
(A) equal to t (B) greater than t
(C) less than t (D) none of these

6. In a mixture of gases, the average number of degree of freedom per molecule is 6. The rms speed of the molecules of the gas is C. The velocity of sound in the gas is
(A) $\frac{C}{\sqrt{2}}$ (B) $\frac{3C}{4}$
(C) $\frac{2C}{3}$ (D) $\frac{C}{\sqrt{3}}$

7. A wall of width ℓ and cross sectional area A is having variable coefficient of thermal conductivity given by $k = k_0 + \alpha x$ (where k_0 and α are positive constant) and x is measured from outer surface of wall. If the temperature of surroundings is T_0°C and the temperature of the room is maintained at T_r °C($T_0 > T_r$). The temperature of the wall at $x = \ell/2$ is (given $\alpha\ell/K_0 = 1$ S.I. unit, $T_0 = 45$°C, and $T_r = 20$°C)
(A) 20°C (B) 25°C
(C) 28°C (D) 30.37°C

8. A particle of mass 2 kg is moving with a uniform speed of $3\sqrt{2}$ m/s in XY plane along the line y = (x + 4)m. The magnitude of the angular momentum about origin is
(A) zero (B) 24 kgm^2/sec
(C) 30 kgm^2/sec (D) 35 kgm^2/sec

9. A bob of mass m is in equilibrium with the help of two inextensible string connected to fixed support. The bob is slightly displaced perpendicular to the plane of figure and released. The time period of oscillation of bob is



(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{d}{g}}$
(C) $2\pi\sqrt{\frac{\ell + d}{g}}$ (D) $2\pi\sqrt{\frac{(\sqrt{4\ell^2 - d^2})}{2g}}$

Space for rough work

Assertion - Reason Type

*This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

10. STATEMENT-1: A ball is projected upwards with such a velocity so as to reach to a height H. Collision of ball with ground is elastic. There is a roof at height h < H. Whatever may be the value of h and e (the coefficient of restitution between the roof and ball), the ball will definitely collide with the roof for second time provided $e \neq 0$.



because

STATEMENT-2: Magnitude of velocity will not change by the collision with ground.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

11. STATEMENT-1: Radius of circular orbit of a satellite is made two times, then its areal velocity will also becomes two times.

because

STATEMENT-2: Areal velocity is given as $\frac{dA}{dt} = \frac{L}{2m} = \frac{mvr}{2m}$.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: If a particle is projected from the ground with some velocity (not very large) at an angle with horizontal, path of the particle is assumed parabolic, although it is not exactly parabolic.

because

STATEMENT-2: Magnitude of acceleration due to gravity at all places is assumed to be the constant, although it is not so.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: The two bodies of masses M and m (M > m) are allowed to fall from the same height. If the air resistance for each be the same then both the bodies will reach the earth simultaneously.

because

STATEMENT-2: For same air resistance, acceleration of both the bodies will be different.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 14 to 16

The air column in a pipe closed at one end is made to vibrate in its third over tone by tuning fork of frequency 220 Hz. The speed of sound in air is 330 m/sec. End correction may be neglected. Let P_0 denote the mean pressure at any point in the pipe and ΔP_0 the maximum amplitude of pressure vibration.

14. Find the length of the air column
(A) 3.2 m (B) 2.625 m
(C) 4.23 m (D) 1.16 m

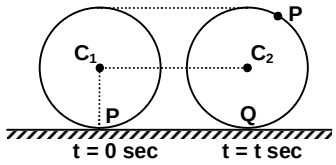
15. What is the amplitude of pressure variation at the middle of the column
(A) $\frac{\Delta P_0}{\sqrt{2}}$ (B) $\frac{\sqrt{3}\Delta P_0}{2}$
(C) ΔP_0 (D) $\frac{\Delta P_0}{2}$

16. What is the maximum pressure at the open end of the pipe?
(A) P_0 (B) $P_0 + \Delta P_0$
(C) $P_0 + \frac{P_0}{2}$ (D) none of these

Space for rough work

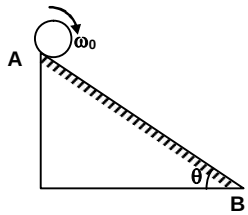
Paragraph for Question Nos. 17 to 19

Rolling motion can be considered as superposition of pure translation and pure rotation about centre of mass of a body of uniform density. Rolling without slipping is a special case of rolling, in which the distance travelled by centre of mass of body of uniform density is equal to the distance travelled by point of contact on body of uniform density [say P] in a given time interval.



If $C_1C_2 = \overline{PQ} \Rightarrow$ Rolling without slipping, and if $C_1C_2 \neq \overline{PQ} \Rightarrow$ Rolling with slipping

Hence if velocity and tangential acceleration of point P with respect to surface must be zero but point P has a total acceleration in case of rolling without slipping. The rolling without slipping can also be viewed as a pure rotation about an axis passing through the point of contact and perpendicular to plane of motion, called "Instantaneous axis of rotation



A disc having radius R and mass M is placed gently on rough inclined surface as shown in the figure. Disc rolls down and comes to B in time t after placing at A. If AB is ℓ and ω_0 is the angular velocity of disc in clockwise direction just before placing it at A (Assume that the friction is sufficient to provide pure rolling to the disc for the given length of inclined plane).

17. A body performing rolling with slipping then
    (A) the radial velocity of point of contact of body relative surface is zero
    (B) the tangential velocity of point of contact of body relative to surface is zero
    (C) the net velocity of point of contact of body relative to surface is zero
    (D) none of the above

18. The nature of rolling of disc during motion from A to B is
    (A) Rolling with slipping (B) rolling without slipping
    (C) some time (A) sometime (B) (D) can't say

19. Work done by frictional force during time interval t is
    (A) zero (B) positive
    (C) Negative (D) none of the above

Space for rough work

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Match the processes in column I with the possibilities in column II, where ΔQ and W are heat change and work done respectively.

Column A	Column B
(A) Isobaric	(p) $\Delta Q > O$
(B) Isothermal	(q) $\Delta Q < O$
(C) Isochoric	(r) $W > O$
(D) Adiabatic	(s) $W < O$

Space for rough work

2. If the position vector of a particle at point P moving in space and distance traveled from a fixed point on the path are given by $\vec{r}$ and s respectively. Three vectors are defined as follows

$$\vec{N}=\frac{d\vec{r}}{ds},\ \vec{T}=R\frac{d\vec{N}}{ds} \text{ and } \vec{B}=\vec{N}\times\vec{T}$$

where R is the radius of curvature at point P and $\vec{r}$ is a non-zero vector. Now match the following on the basis of above concept

Column A	Column B
(A) $\vec{T}$	(p) Unit vector
(B) $\vec{N}$	(q) Zero
(C) $\vec{N}\cdot\vec{T}$	(r) $\hat{v}$ (unit vector along the direction of velocity)
(D) $\vec{B}$	(s) $\frac{v\vec{a}-\vec{v}\,a_T}{\lvert\vec{v}\times\vec{a}\rvert}$, where $\vec{a}$ and a_T are acceleration and magnitude of tangential acceleration of particle P.

3. Match the following.

Column – A	Column – B
(A) Equation of continuity (Av = constant)	(p) non-viscous flow
(B) Bernoulli's equation	(q) steady flow
(C) Velocity at a given point is constant with time.	(r) incompressible flow
(D) Ideal flow	(s) irrotational flow

Space for rough work

Chemistry

PART - II

SECTION – A

Single Correct Choice Type

*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. Lattice energy of chloride of alkali metals is in order:
   (A) LiCl > NaCl > KCl > RbCl > CsCl
   (B) LiCl < NaCl < KCl < RbCl < CsCl
   (C) NaCl < KCl < LiCl < RbCl > CsCl
   (D) NaCl < KCl < RbCl < CsCl > LiCl

2. A ball of mass 200 g is moving with a velocity of 10 m/sec. if the error is the measurement of velocity is 0.1%, then the uncertainty in its position is
   (A) 3.3×10^{-31} m
   (B) 3.3×10^{-27} m
   (C) 5.3×10^{-25} m
   (D) 2.6×10^{-32} m

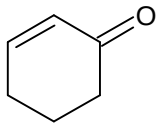
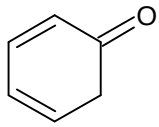
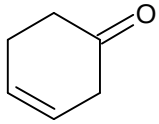
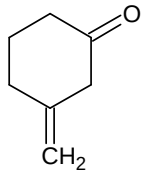
3. Fluorine reacts with uranium to form UF_6:
   $U(s) + 3F_2(g) \longrightarrow UF_6(g)$
   How many fluorine molecules are required to produce 2 mg of UF_6 from an excess of uranium? (The molar mass of UF_6 is 352 g/mol)
   (A) 3.4×10^{18}
   (B) 1×10^{19}
   (C) 2×10^{19}
   (D) 3.4×10^{21}

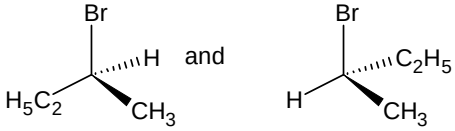
4. In the disproportion reaction:
   $H_3PO_2 \longrightarrow PH_3 + H_3PO_3$
   The equivalent mass of H_3PO_2 is (m = molecular mass of H_3PO_2)
   (A) $\frac{3m}{4}$
   (B) $\frac{3m}{5}$
   (C) $\frac{5m}{6}$
   (D) $\frac{m}{4}$

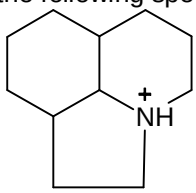
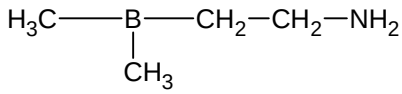
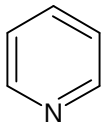
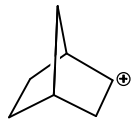
Space for rough work

5. For the reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. $K_p = 6.1 \times 10^{-5}$ at 25°C. ΔG° for the reaction will be ($\log_{10}61 = 1.78$)
(A) 24 kJ (B) −24 kJ
(C) −7.3 kJ (D) 2 kJ

6. A diatomic molecule has a dipole moment of 1.2 D. If the bond length is 1.0×10^{-8} cm, what fraction of charge does exist on each atom?
(A) 0.1 (B) 0.2
(C) 0.25 (D) 0.3

7. Which of the following has longest C – O bond?
(A)  (B) 
(C)  (D) 

8. The given pair is

(A) enantiomers (B) homomers
(C) constitutional isomers (D) diastereomers

9. Which of the following species will act only as electrophile and not as nucleophile?
(A)  (B) 
(C)  (D) 

Space for rough work

Assertion - Reason Type

This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ***ONLY ONE*** *is correct.*

10. STATEMENT-1: In Lother Meyer Curve the alkali elements occupy maxima of the curve.
and
STATEMENT-2: Alkali elements have largest volume in a period.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

11. STATEMENT-1: Cyclopropane has greater ring strain than cyclobutane ,yet this is readily formed.
and
STATEMENT-2: Formation of four membered ring involves larger decrease in entropy than that of three membered ring.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: pH of a neutral solution is always equal to $p^{k_w}/2$ at any temperature.
and
STATEMENT-2: A neutral solution at any temperature will have same concentration of $\left[H^+\right]$ and $\left[OH^-\right]$ ion in the expression for K_w.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

13. STATEMENT-1: 0.74g of $Ca(OH)_2$ is required for 10 litre of water to remove temporary hardness of 100 ppm due to $Mg(HCO_3)_2$ (atomic mass: Ca = 40, O = 16, H = 1,Mg = 24, C = 12).
and
STATEMENT-2: Hardness of water is caused due to calcium and magnesium salts of Cl^-, SO_4^{2-} and HCO_3^-.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 14 to 16

Alkene undergo free radical substitution reaction as it is carried out in the presence of sunlight. Experimentally it is found that relative rate of abstraction of hydrogen atoms from 1°, 2° and 3° carbon atoms are in the ratio 1: 3.8 : 5 during the monobromination while this ratio is 1 ; 82 : 1600 during the monobromination. Alkyl halide undergoes nucleophilic substituton reaction which may be unimolecular or bimolecular depending upon the nature of substrate alkyl halide. Former reaction proceeds via the carbocation intermediate while latter one proceeds via transition state.

14. $CH_2 = CH - CH_2 - CH_3 \xrightarrow{Cl_2/\Delta} (A) + (B)$

In the given equation (A) and (B) are

(A) $H_2C{=}CH{-}\underset{\displaystyle Cl}{\underset{|}{CH}}{-}CH_3$ and $H_2C{=}CH{-}CH_2{-}\underset{\displaystyle Cl}{\underset{|}{CH_2}}$

(B $H_2C{-}\underset{\displaystyle Cl}{\underset{|}{CH}}{=}CH{-}CH_3$ and $H_2C{=}CH{-}\underset{\displaystyle Cl}{\underset{|}{CH}}{-}CH_3$

(C) $H_2C{-}\underset{\displaystyle Cl}{\underset{|}{CH}}{-}CH_2{-}CH_3$ and $HC{=}\underset{\displaystyle Cl}{\underset{|}{CH}}{-}CH_2{-}CH_3$

(D) $H_2C{=}\underset{\displaystyle Cl}{\underset{|}{C}}{-}CH_2{-}CH_3$ and $H_2C{=}CH{-}\underset{\displaystyle Cl}{\underset{|}{CH}}{-}CH_3$

15. Predict the product ratio during the monochlorination of 2 – methylbutane?
(A) 9 : 5 : 3 (B) 9 : 5 : 3.8 : 3
(C) 9 : 7.6 : 5 (D) 6 : 3 : 7.6 : 5

16. % isomer of products corresponding to 1° and 3° C – atoms obtained during the monobromination of 2 – methylpropane are
(A) 0.5% and 99.5% (B) 80% and 20%
(C) 75% and 25% (D) 15% and 85%

Space for rough work

Paragraph for Question Nos. 17 to 19

Werner Heisenberg stated the uncertainty principle which is the consequence of dual behaviour of matter and radiation. It states that if is impossible to determine simultaneously, the exact position and exact momentum (or velocity) of an electron.
Mathematically, it can be given as the equation:

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

Where Δx is the uncertainty in position and Δp_x (or ΔV_x) is the uncertainty in momentum (or velocity) of the particle.
If the position of the electron known with high degree of accuracy (Δx is small), then the velocity of the electron will be uncertain [$\Delta(V_x)$ is large]. On the other hand, if the velocity of the electron is known precisely [$\Delta(V_x)$ is small], then the position of the electron will be uncertain (Δx will be large). Thus if we carry out some physical measurements on the electron's position or velocity, the outcome will always depict a fuzzy or blue picture.

17. An electron moves around the nucleus in a circle of radius r. Assuming that the uncertainty of the momentum of electron is of the same order as momentum itself, the momentum of the electron would be

(A) $\frac{h}{4\pi r}$ (B) $2\pi hr$

(C) $\frac{h}{2\pi r}$ (D) $4\pi hr$

18. The uncertainty in the velocity of a cricket ball of mass 100 g, when uncertainty in its position is of the order of 1A°, would be

(A) $\geq 5.27 \times 10^{-24} ms^{-1}$ (B) $\geq 2.35 \times 10^{-23} ms^{-1}$

(C) $\geq 3.16 \times 10^{-22} ms^{-1}$ (D) $\geq 8.51 \times 10^{-24} ms^{-1}$

19. If one tries to find the exact location of an electron to an uncertainty of only 10^{-8} m, then uncertainty in velocity, i.e. ΔV would be

(A) 10^{-4} ms^{-1} (B) 10^{4} ms^{-1}

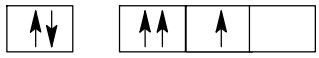
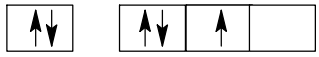
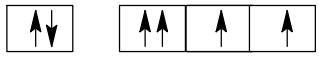
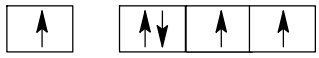
(C) 10^{-8} ms^{-1} (D) 10^{8} ms^{-1}

Space for rough work

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

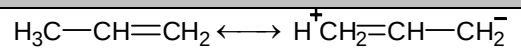
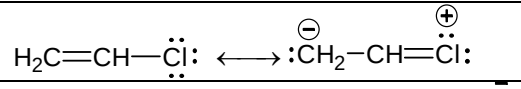
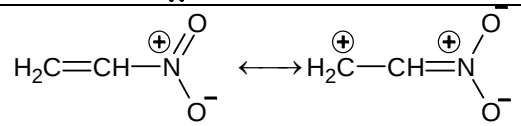
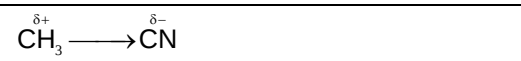
1. Match the electronic configuration listed in Column – II with the descriptions listed in Column – I:

Column – I		Column – II	
(A)	Violation of Aufbau's rule	(p)	
(B)	Violation of Pauli's exclusion principle	(q)	
(C)	Violation of Hund's rule	(r)	
(D)	Violation of both Pauli's and Hund's rules	(s)	

2. Match the following Column – I with Column – II:

Column – I		Column – II	
(A)	C_2	(p)	$\sigma 2p_Z$ ($\pi 2p_x = \pi 2p_y$)
(B)	O_2	(q)	Paramagnetic
(C)	NO^+	(r)	Triple bond
(D)	N_2	(s)	Diamagnetic

3. Match the following Column – I with Column – II:

Column – I		Column – II	
(A)		(p)	–I effect
(B)		(q)	–R effect
(C)		(r)	+R effect
(D)		(s)	Hyper conjugation effect

Space for rough work

Mathematics PART – III

SECTION – A

Single Correct Choice Type

*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. Consider the sequence 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, ….. then 1025th term will be
(A) 2^9 (B) 2^{11}
(C) 2^{10} (D) 2^{12}

2. A student is allowed to select at most n books from a collection of (2n + 1) books. If the total number of ways in which he can select at least one book is 63, then the value of n is
(A) 2 (B) 3
(C) 4 (D) 5

3. If $|z - 1| \le 2$ and $|\omega z - 1 - \omega^2| = a$ (where ω is cube root of unity) then complete set of values of a is
(A) $0 \le a \le 2$ (B) $\frac{1}{2} \le a \le \frac{\sqrt{3}}{2}$
(C) $\frac{\sqrt{3}}{2} - \frac{1}{2} \le a \le \frac{1}{2} + \frac{\sqrt{3}}{2}$ (D) $0 \le a \le 4$

4. In triangle ABC, a = 5, b = 4 and c = 3. G is the centroid of the triangle. Circumradius of triangle GAB is equal to
(A) $2\sqrt{13}$ (B) $\frac{5}{12}\sqrt{13}$
(C) $\frac{5}{3}\sqrt{13}$ (D) $\frac{3}{2}\sqrt{13}$

Space for rough work

5. If $x \in [-1, 0)$, then $\cos^{-1}(2x^2 - 1) - 2\sin^{-1} x$ is equal to

(A) $-\frac{\pi}{2}$ (B) π

(C) $\frac{3\pi}{2}$ (D) -2π

6. The range of values of α for which the line $2y = gx + \alpha$ is a normal to the circle $x^2 + y^2 + 2gx + 2gy - 2 = 0$ for all values of g is

(A) $[1, \infty)$ (B) $[-1, \infty)$

(C) $(0, 1)$ (D) $(-\infty, 1]$

7. If $y = m_1x + c$ and $y = m_2x + c$ are two tangents to the parabola $y^2 + 4a(x + a) = 0$, then

(A) $m_1 + m_2 = 0$ (B) $1 + m_1 + m_2 = 0$

(C) $m_1m_2 - 1 = 0$ (D) $1 + m_1m_2 = 0$

8. $(x - 1)(y - 2) = 5$ and $(x - 1)^2 + (y + 2)^2 = r^2$ intersect at four points A, B, C, D and if centroid of ΔABC lies on line $y = 3x - 4$, then locus of D is

(A) $y = 3x$ (B) $x^2 + y^2 + 3x + 1 = 0$

(C) $3y = x + 1$ (D) $y = 3x + 1$

9. If roots of an equation $x^n - 1 = 0$ are $1, a_1, a_2, \ldots, a_{n-1}$, then the value of $(1 - a_1)(1 - a_2)(1 - a_3) \ldots (1 - a_{n-1})$ will be

(A) n (B) n^2

(C) n^n (D) 0

Assertion - Reason Type

This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ***ONLY ONE*** *is correct.*

10. STATEMENT 1: 1111 ..... 91 times is composite number
STATEMENT 2: 91 is composite number
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

Space for rough work

11. STATEMENT 1: If n is an odd integer greater than 3 but not a multiple of 3, then $(x + 1)^n - x^n - 1$ is divisible by $x^3 + x^2 + x$
STATEMENT 2: If n is an odd integer greater than 3 but not a multiple of 3, we have $1 + \omega^n + \omega^{2n} = 3$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

12. STATEMENT 1: The minimum value of $27^{\cos 2x}81^{\sin 2x}$ is $\frac{1}{243}$
STATEMENT 2: The minimum value of $a \cos \theta + b \sin \theta$ is $-\sqrt{a^2 + b^2}$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

13. STATEMENT 1: The locus of a moving point satisfying $\sqrt{(x+2)^2 + y^2} + \sqrt{(x-2)^2 + y^2} = 4$ is ellipse
STATEMENT 2: Distance between (–2, 0) and (2, 0) is 4
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

Space for rough work

Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:

A(1, 3) and $C\left(-\frac{2}{5}, -\frac{2}{5}\right)$ are the vertices of a triangle ABC and the equation of the internal angle bisector of $\angle ABC$ is $x + y = 2$

14. Equation of side BC is
(A) $7x + 3y - 4 = 0$ (B) $7x + 3y + 4 = 0$
(C) $7x - 3y + 4 = 0$ (D) $7x - 3y - 4 = 0$

15. Co-ordinates of vertex B are
(A) $\left(\frac{3}{10}, \frac{17}{10}\right)$ (B) $\left(\frac{17}{10}, \frac{3}{10}\right)$
(C) $\left(-\frac{5}{2}, \frac{9}{2}\right)$ (D) (1, 1)

16. Equation of side AB is
(A) $3x + 7y = 24$ (B) $3x + 7y + 24 = 0$
(C) $13x + 7y + 8 = 0$ (D) $13x - 7y + 8 = 0$

Space for rough work

Paragraph for Question Nos. 17 to 19

Read the following write up carefully and answer the following questions:
Five balls are to be placed in three boxes. Each box should hold all the five balls so that no box remains empty

17. Number of ways if balls are different but boxes are identical is
(A) 30 (B) 25
(C) 21 (D) 35

18. Number of ways if balls and boxes are identical is
(A) 3 (B) 1
(C) 2 (D) none of these

19. Number of ways if balls as well as boxes are identical but boxes are kept in a row is
(A) 10 (B) 15
(C) 20 (D) 6

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	If a, b, c are in G.P., then $\log_a 10$, $\log_b 10$, $\log_c 10$ are in	(p)	A.P.
(B)	If $\frac{a+be^x}{a-be^x}=\frac{b+ce^x}{b-ce^x}=\frac{c+de^x}{c-de^x}$, then a, b, c, d are in	(q)	H.P.
(C)	If a, b, c are in A.P., a, x, b are in G.P. and b, y, c are in G.P., then x^2, b^2, y^2 are in	(r)	G.P.
(D)	If x, y, z are in G.P., $a^x = b^y = c^z$, then log a, log b, log c are in	(s)	None of these

Space for rough work

2. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	$\cot\frac{A}{2}=\frac{b+c}{a}$	(p)	always right angled
(B)	$a\tan A + b\tan B = (a+b)\tan\left(\frac{A+B}{2}\right)$	(q)	always isosceles
(C)	$a\cos A = b\cos B$	(r)	may be right angled
(D)	$\cos A = \frac{\sin B}{2\sin C}$	(s)	may be right angled isosceles

3. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	If two circles $x^2 + y^2 + 2a_1x + b = 0$ and $x^2 + y^2 + 2a_2x + b = 0$ touch each other then triplet (a_1, a_2, b) can be	(p)	(2, 2, 2)
(B)	If two circles $x^2 + y^2 + 2a_1x + b = 0$ and $x^2 + y^2 + 2a_2y + b = 0$ touch each other then triplet (a_1, a_2, b) can be	(q)	$\left(1, 1, \frac{1}{2}\right)$
(C)	If the straight line $a_1x - by + b^2 = 0$ touches the circle $x^2 + y^2 = a_2x + by$, then triplet (a_1, a_2, b) can be	(r)	(2, 1, 0)
(D)	If the line $3x + 4y - 4 = 0$ touches the circle $(x - a_1)^2 + (y - a_2)^2 = b^2$, then triplet (a_1, a_2, b) can be	(s)	$\left(1, 1, \frac{3}{5}\right)$

Space for rough work